

Public Transportation Locator System – System Planning & Feasibility Report (Nepal)

1. Introduction

1.1 Background

Public transportation in Nepal is largely semi-formal and fragmented. While buses, microbuses, tempos, magic vans, and jeeps form the backbone of daily mobility, passengers face major problems: - Uncertainty about vehicle arrival times - Lack of route clarity - No unified real-time tracking system - Poor access to information in low-internet regions

This project proposes a **Public Transportation Locator System** (mobile app + web app), designed **specifically for Nepal**, focusing on **low-connectivity environments**, **informal transport**, and **cost-effective tracking**.

1.2 Vision

To create a **reliable, low-cost, offline-friendly public transport tracking platform** that benefits commuters, transport operators, municipalities, and private stakeholders.

2. System Objectives

- Provide near real-time visibility of public transport vehicles
 - Function reliably in **low or no internet conditions**
 - Support **informal and formal transport types**
 - Minimize hardware dependency
 - Enable future scalability across Nepal
 - Create sustainable monetization opportunities
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3. Stakeholder Analysis

3.1 Primary Stakeholders

Stakeholder	Needs	Opportunities
Daily Commuters	ETA, route clarity, trust	Time saving, safer travel
Drivers / Conductors	More passengers, ease of use	Higher occupancy, incentives
Transport Companies	Route analytics, visibility	Data-driven optimization
Municipalities	Traffic insight, planning	Smart city integration

3.2 Secondary Stakeholders

Stakeholder	Role
ISPs (WorldLink, etc.)	Connectivity & hotspots
Advertisers	Revenue generation
NGOs / Mapping Communities	Data validation
Hardware Vendors	Display & signage

4. Key Challenges & Constraints

4.1 Connectivity Constraints

- Unreliable mobile data
- High cost of continuous internet usage
- Rural and highway coverage gaps

4.2 GPS & Tracking Challenges

- GPS disabled by drivers
- Battery drain concerns
- Urban signal distortion
- No standardized vehicle hardware

4.3 Data Challenges

- No official GTFS feeds
- Route changes without notice
- Informal and flexible routing

4.4 Human & Operational Challenges

- Driver resistance
- Low digital literacy
- Trust and adoption barriers

4.5 Legal & Privacy Constraints

- User location sensitivity
 - Data ownership
 - Consent requirements
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5. Proposed Multi-Layer Solution Architecture

5.1 Design Philosophy

- **Offline-first**

- **Redundancy-based tracking**
 - **Prediction over precision**
 - **Transparency over false accuracy**
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6. Tracking & Data Collection Strategies

6.1 Primary Layer: Driver Phone GPS (Offline-First)

- GPS points stored locally
- Periodic burst sync when network available
- Route-locked prediction between syncs

Advantages: - High accuracy - Low cost - Scalable

Limitations: - Driver cooperation required

6.2 Secondary Layer: Cell Tower Approximation

- Uses Cell ID & signal strength
- Works without GPS
- Accuracy: 200–2000m

Use Cases: - Highways - Rural areas - GPS failure fallback

6.3 Tertiary Layer: Crowd-Sourced Passenger Detection

- Cluster passengers moving together
- Speed & direction matching
- Voting-based vehicle type detection

Strengths: - Works without driver app - Covers informal transport

6.4 Advanced Layer: BLE Mesh Networking

- Bluetooth Low Energy communication between passenger phones
- No internet required for most devices
- One internet-enabled device uploads group data

Key Benefit:

One connected phone can relay data for many offline phones

6.5 Fallback Layer: Landmark & SMS Check-Ins

- Manual driver input
- SMS-based updates

- QR / NFC checkpoints

Highly reliable in low-tech environments

7. Offline & Low-Connectivity Strategy

Method	Internet Required	Accuracy	Reliability
Driver GPS (sync later)	Periodic	High	High
Cell Tower	No	Medium	Medium
BLE Mesh	One device	Medium-High	Medium
Landmark / SMS	No	Medium	High

8. System Confidence Model

Instead of exact locations, system shows: - Last confirmed time - Estimated position - Confidence percentage

Example:

Bus near Kalanki – 78% confidence (last confirmed 2 min ago)

9. Basic Graph-Based Design (Conceptual)

9.1 Route Graph

- Nodes = Stops / Landmarks
- Edges = Road segments
- Weights = Time, congestion, reliability

9.2 Vehicle State Graph

STOPPED → STARTED → MOVING → DELAYED → ENDED

9.3 Crowd Consensus Graph

Users → Cluster → Vehicle Identity → Route Match

10. Data Integrity & Validation

- Multi-source confirmation
- Historical consistency checks
- Anomaly detection

- User reputation weighting
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11. Privacy & Ethics Considerations

- No phone number storage
 - Anonymized device IDs
 - Explicit consent
 - Opt-out anytime
 - Aggregated data only
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12. Monetization & Sustainability (High-Level)

- Operator subscriptions
 - Station display advertising
 - Sponsored routes
 - Municipal data services
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13. Risk Assessment

Risk	Impact	Mitigation
Driver non-cooperation	High	Incentives
ISP dependency	Medium	Multi-network
Vandalism (displays)	High	Controlled locations
Data inaccuracy	Medium	Confidence model

14. Phased Implementation Roadmap

Phase 1 – MVP

- Single city (Kathmandu)
- Driver app + public app
- Offline GPS sync

Phase 2 – Intelligence Layer

- Crowd clustering
- Cell tower fallback
- ETA prediction

Phase 3 – Expansion

- BLE mesh

- Public displays
 - Multi-city rollout
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15. Conclusion

This system is **not a typical transit app**. It is a **resilient, Nepal-first mobility intelligence platform** designed for real-world constraints. By combining **offline tracking, crowd intelligence**, and **predictive modeling**, it creates a realistic path toward smarter public transport in Nepal.

This document serves as a system planning and requirement analysis foundation. Detailed algorithms, data flows, technology stack, and implementation specifications can be developed in subsequent phases.